

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of canopy height per plot (esy4 = 4qm plots)

MB

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
```

```
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
```

```
library(ggbeeswarm)
library(patchwork)
library(DHARMA)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.height) %>%
filter(y < 1) # see section Outliers: Exclude site N_DAM (more or less only the tall grass Arrhenathera)
```

Load data

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `esy4 = fct_relevel(esy4, "R", "R22", "R1A")`.
## Caused by warning:
## ! 1 unknown level in `f`: R
```

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 0.4634648 0.4485813 0.4336978
```

```
median(sites$y)
```

```
## [1] 0.44
```

```
sd(sites$y)
```

```
## [1] 0.1285512
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%    95%  
## 0.228 0.686
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2  
##   eco.id     n  
##   <fct> <int>  
## 1 654     101  
## 2 664      89  
## 3 686      99
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2  
##   site.type     n  
##   <ord>    <int>  
## 1 positive     45  
## 2 restored    219  
## 3 negative     25
```

```
sites %>% count(esy4)
```

```
## # A tibble: 2 x 2  
##   esy4     n  
##   <fct> <int>  
## 1 R22    208  
## 2 R1A     81
```

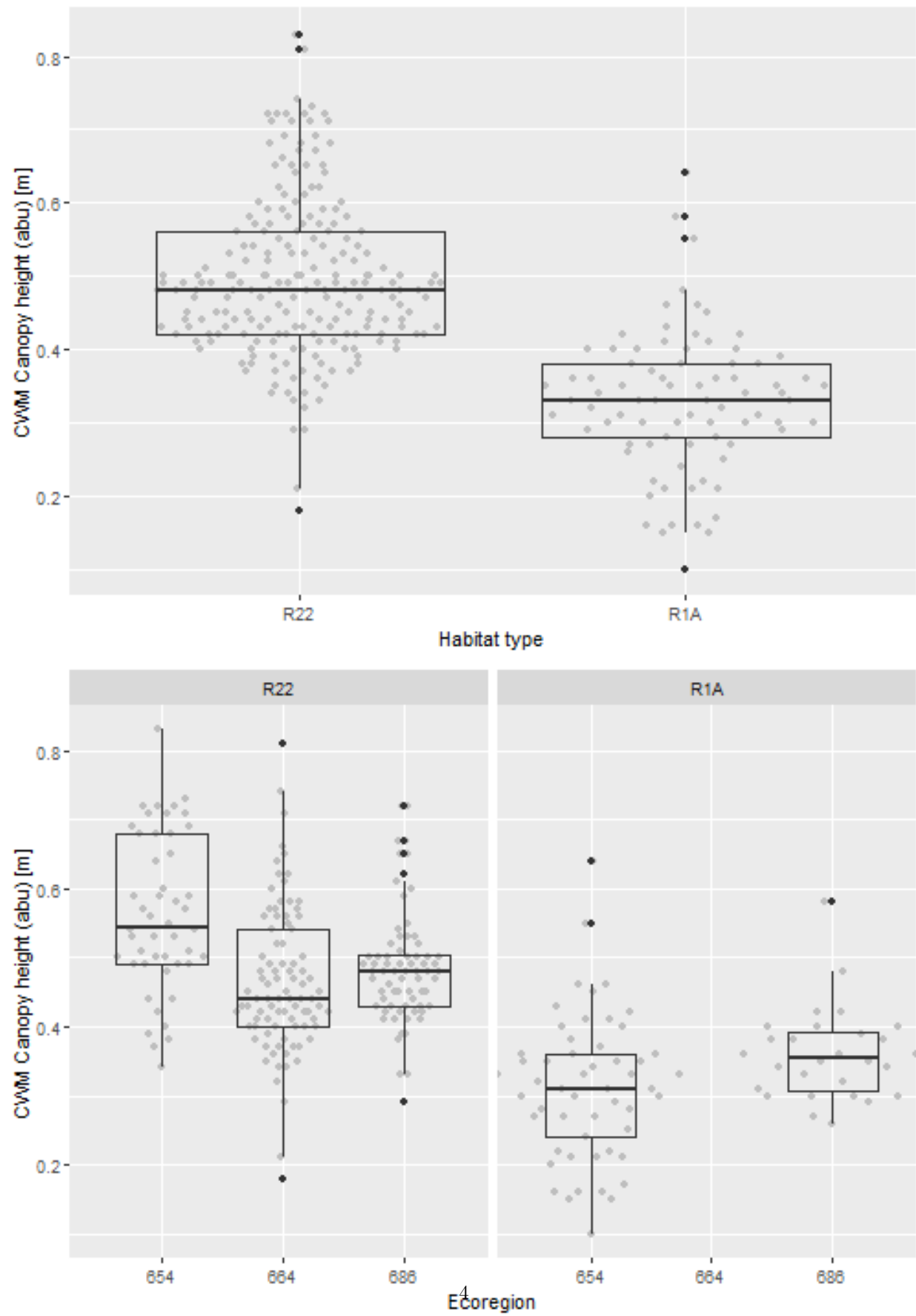
```
sites %>% count(esy4, eco.id)
```

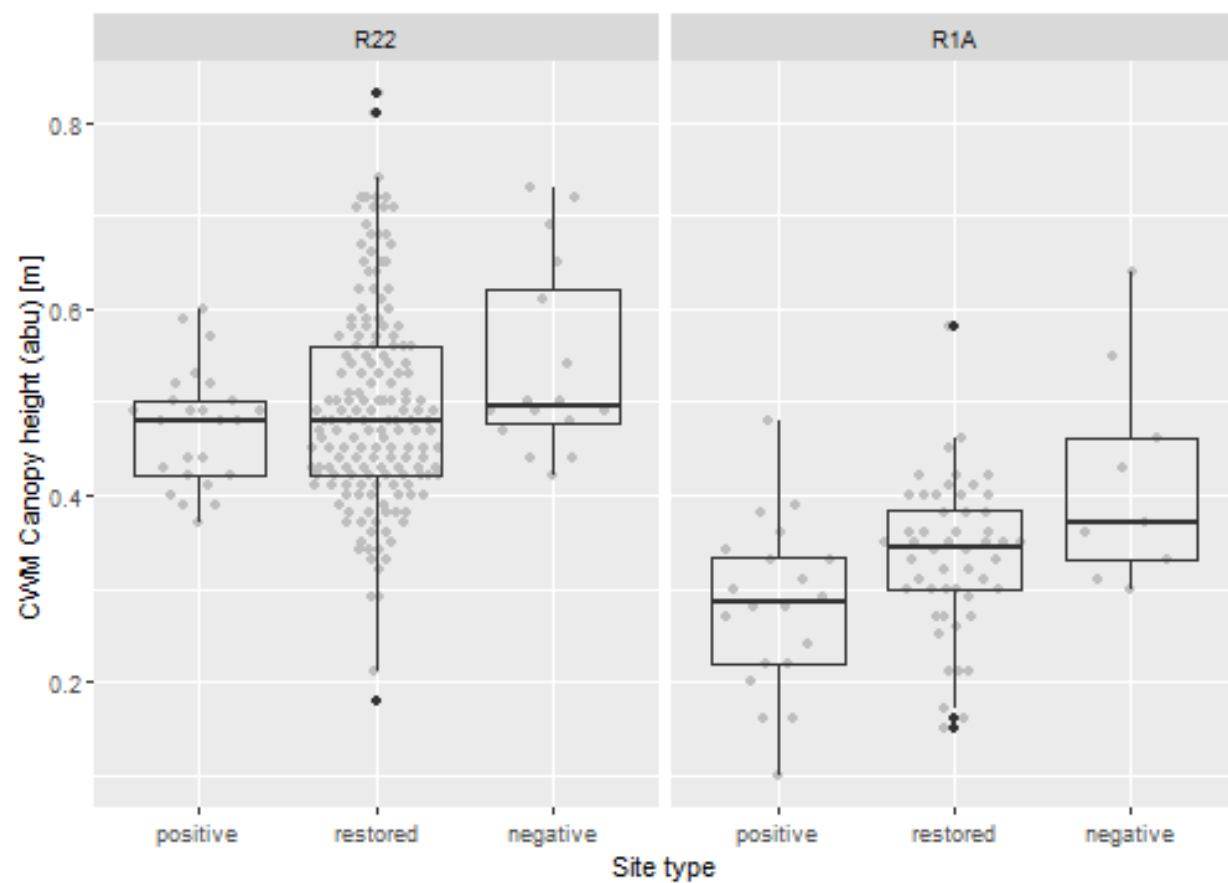
```
## # A tibble: 5 x 3  
##   esy4 eco.id     n  
##   <fct> <fct> <int>  
## 1 R22  654     48  
## 2 R22  664     89  
## 3 R22  686     71  
## 4 R1A  654     53  
## 5 R1A  686     28
```

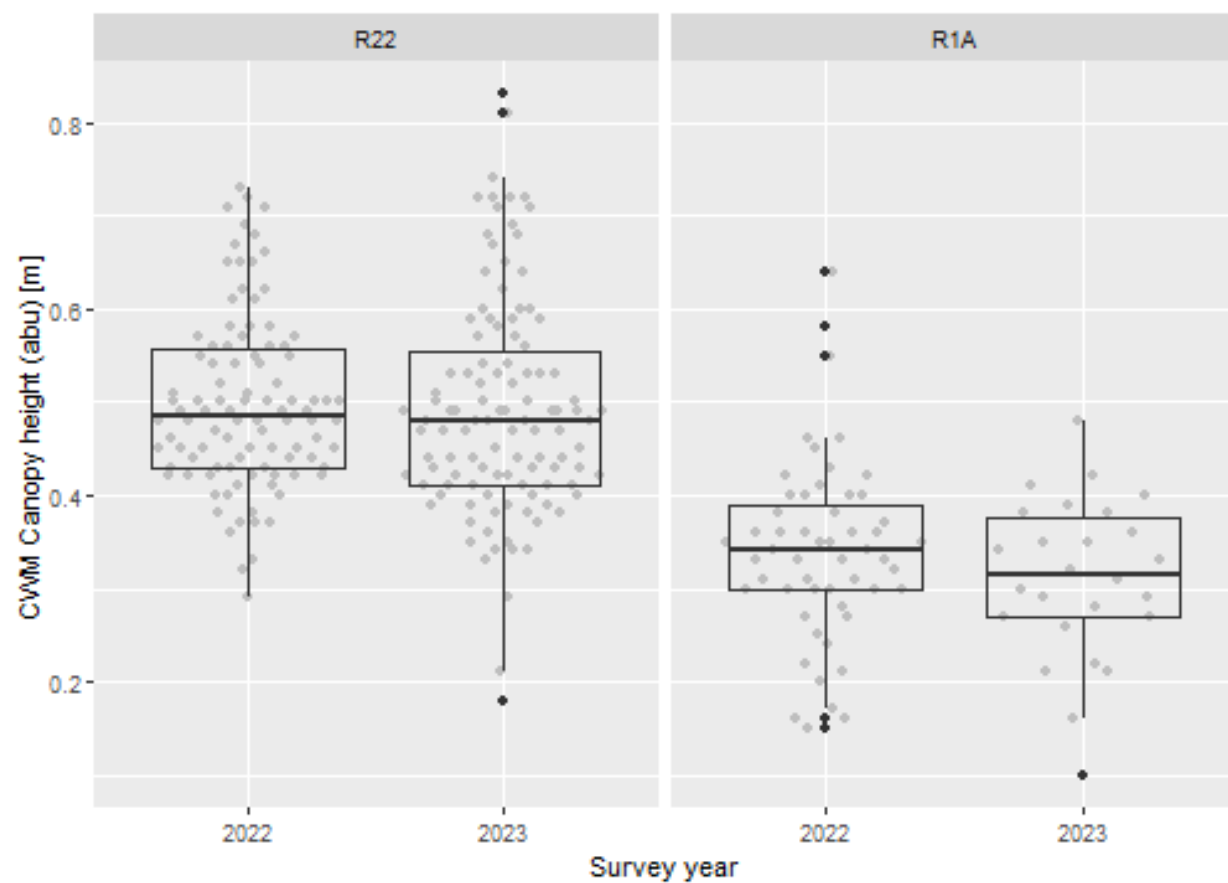
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 6 x 3  
##   esy4 site.type     n  
##   <fct> <ord>    <int>  
## 1 R22  positive     25  
## 2 R22  restored    167  
## 3 R22  negative     16  
## 4 R1A  positive     20  
## 5 R1A  restored     52  
## 6 R1A  negative      9
```

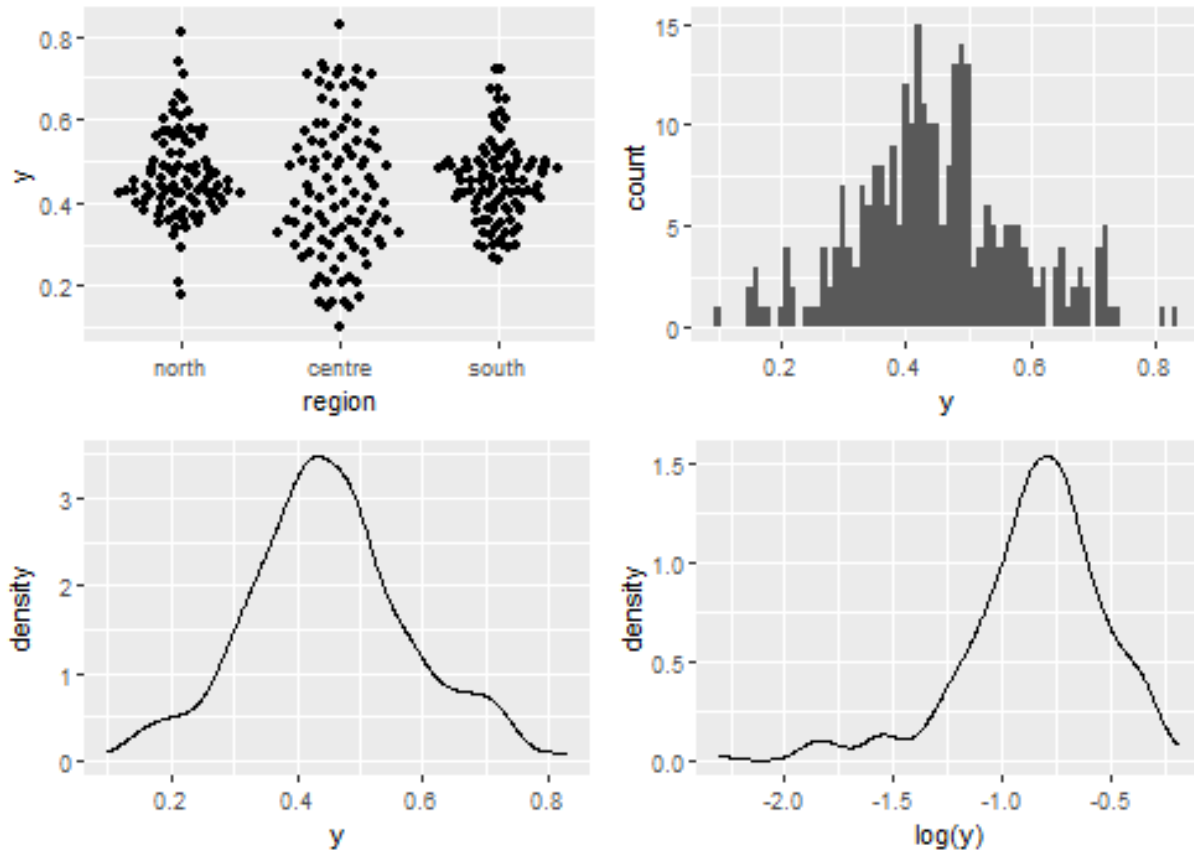
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 *Ecography* DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum.)) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_height_cwm_1.Rdata"))
load(file = here("outputs", "models", "model_height_cwm_2.Rdata"))
m_1 <- m1
m_2 <- m2

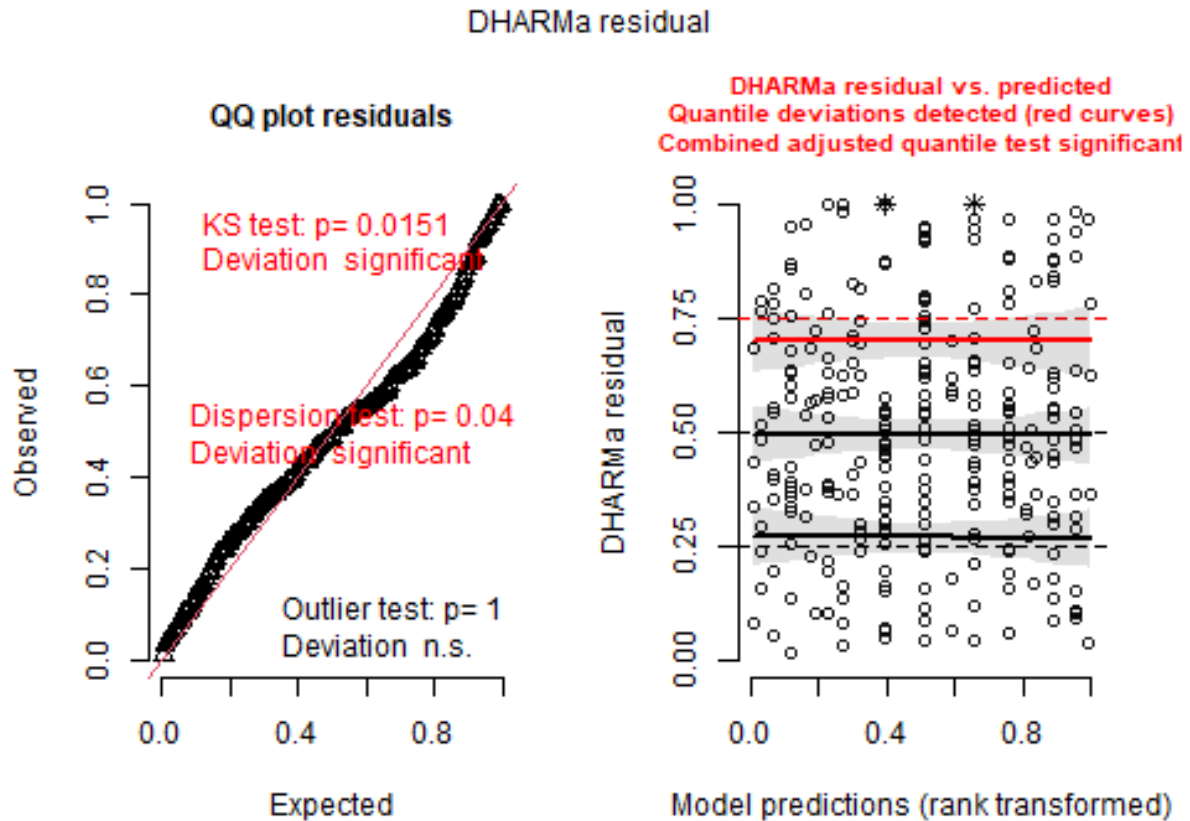
m_1@call
## lmer(formula = y ~ esy4 * (site.type + eco.id) + obs.year + (1 |
##   id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = y ~ (esy4 + site.type + eco.id + obs.year)^2 +
##       (1 | id.site), data = sites, REML = FALSE)
```

Model check

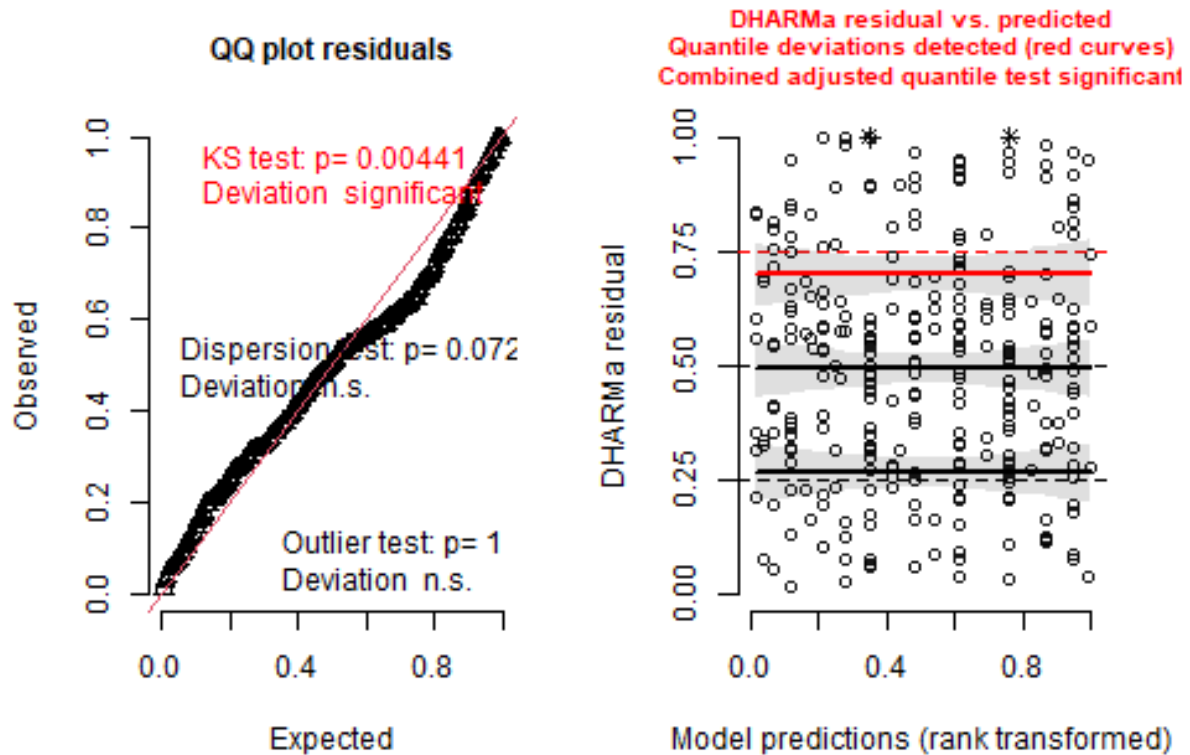
DHARMa

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

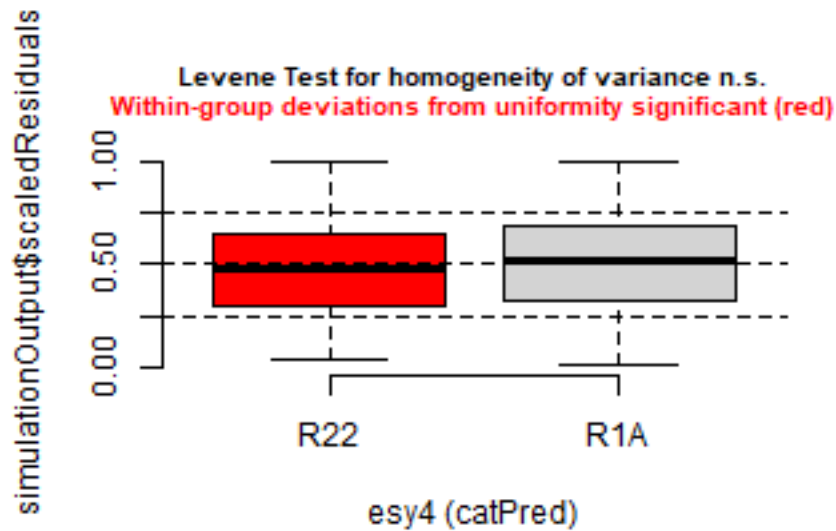


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

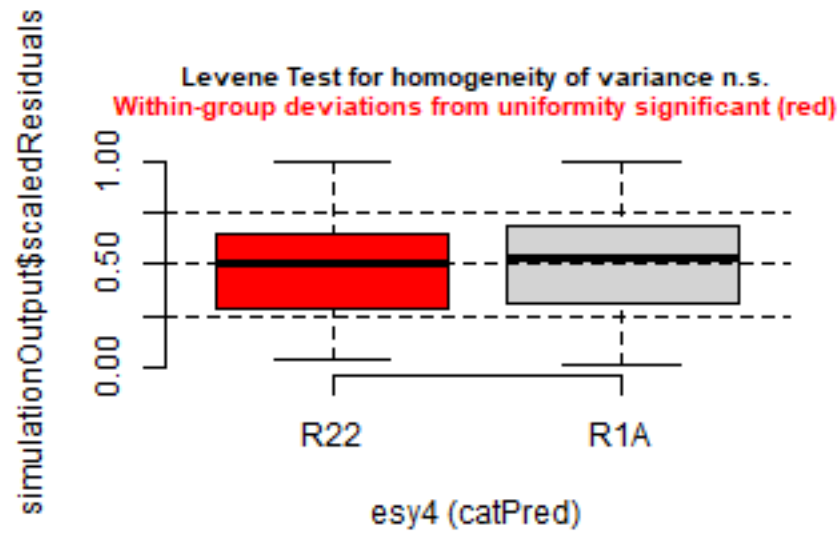

DHARMa residual



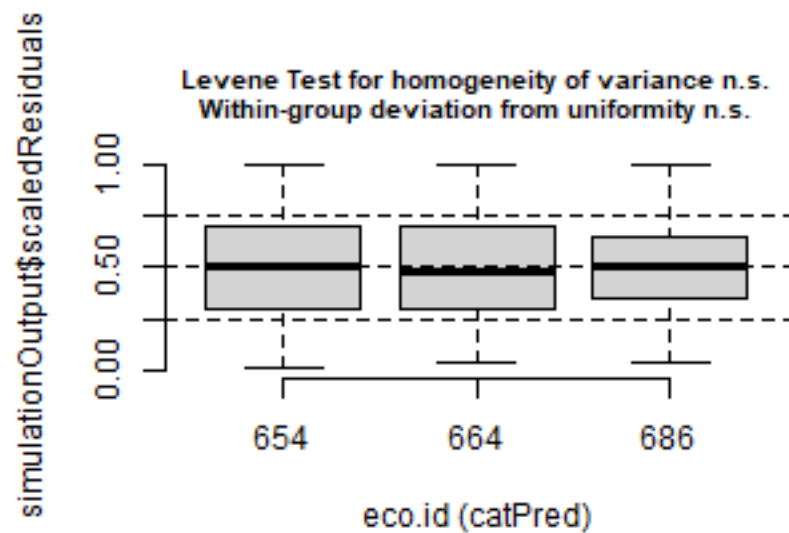
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



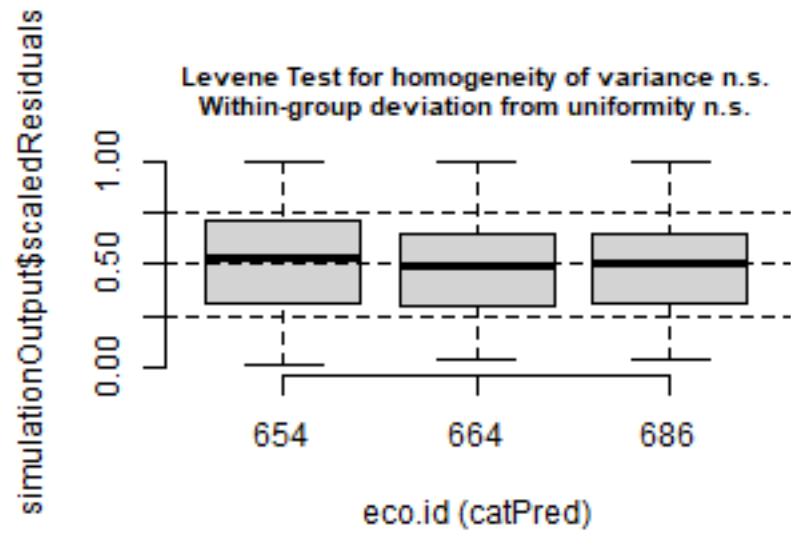
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



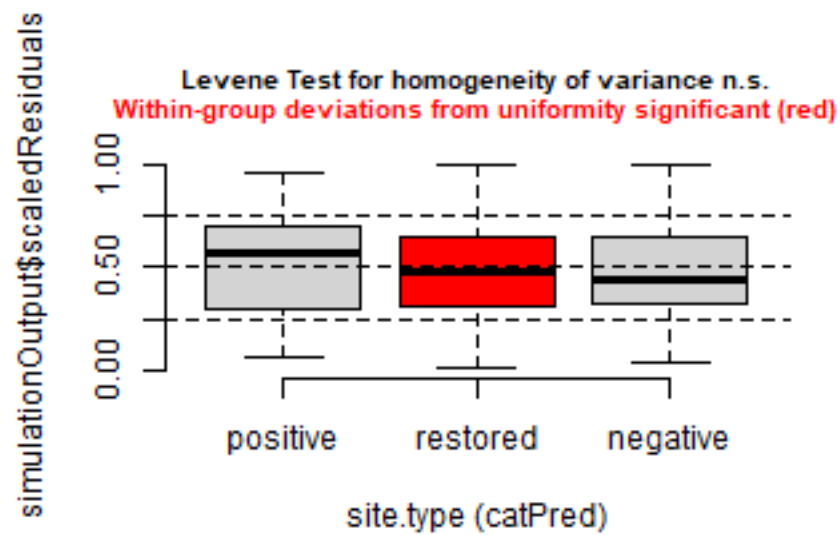
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



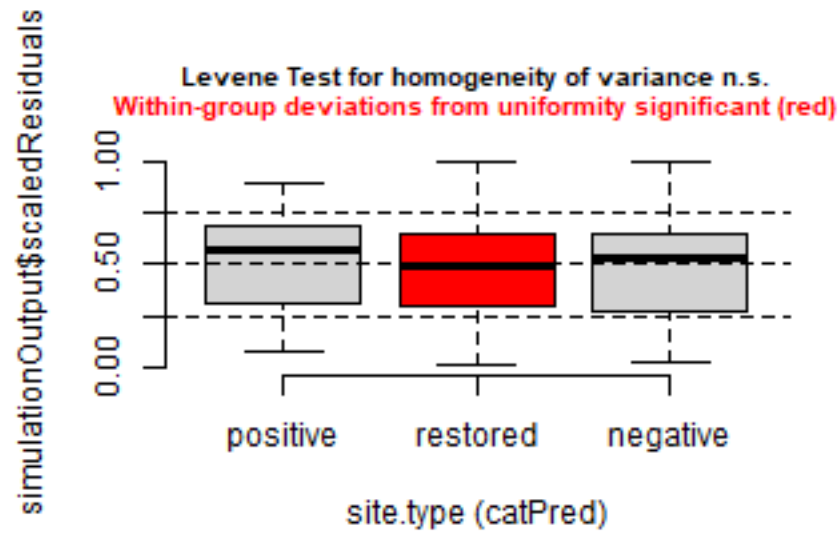
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



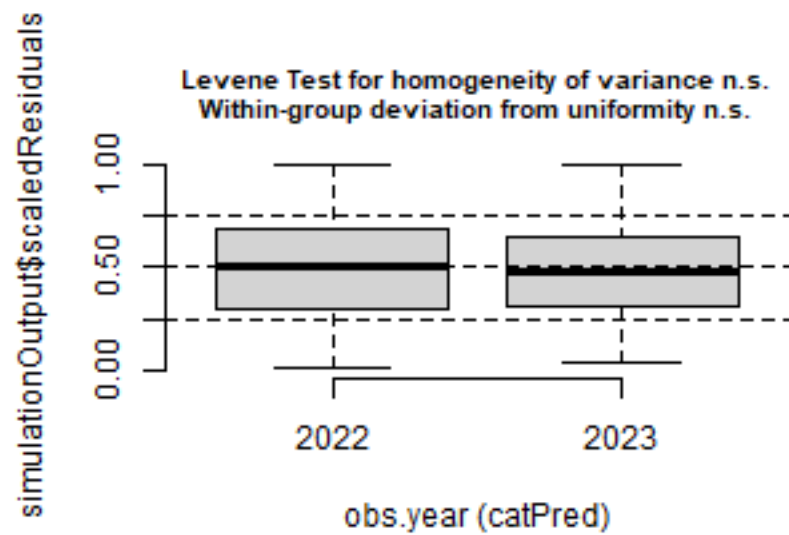
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



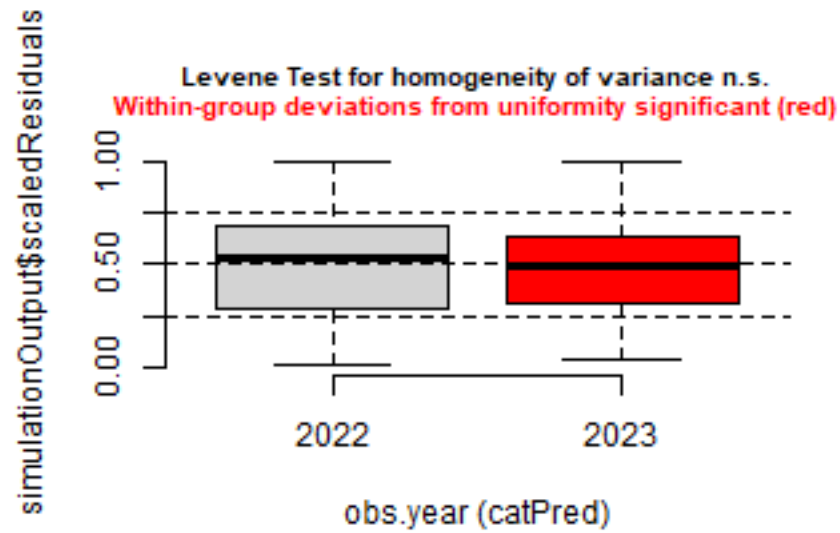
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



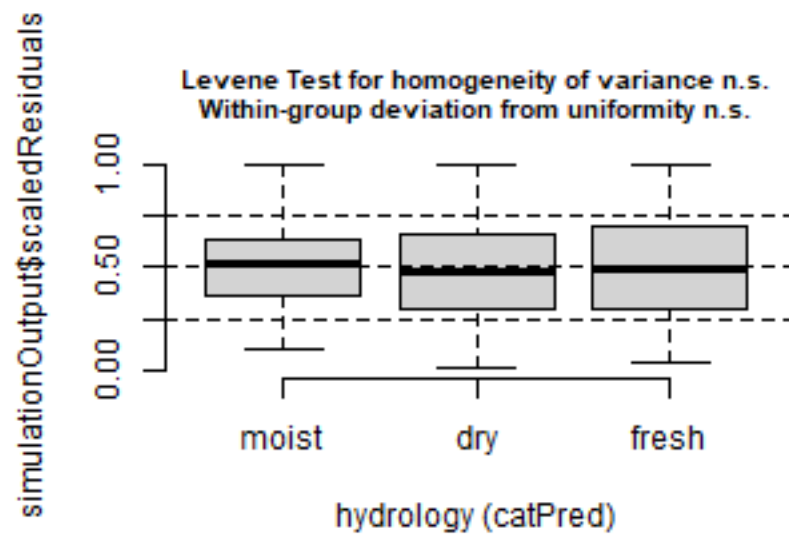
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



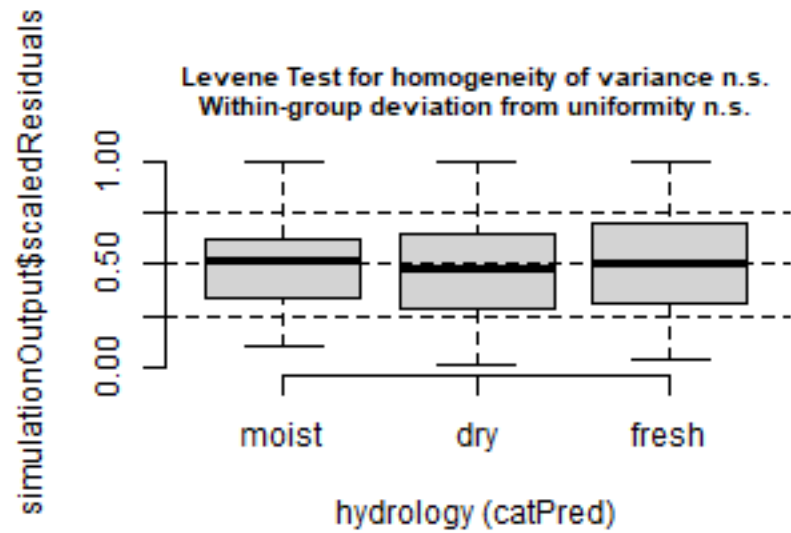
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



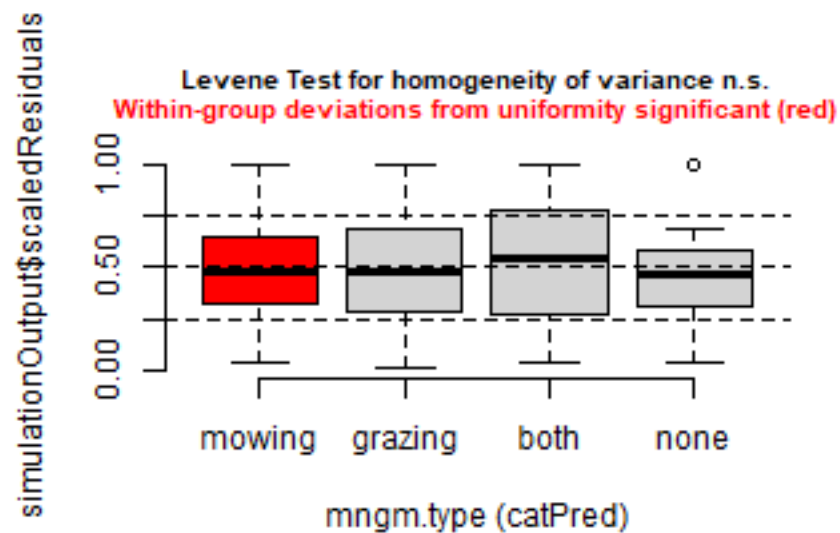
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



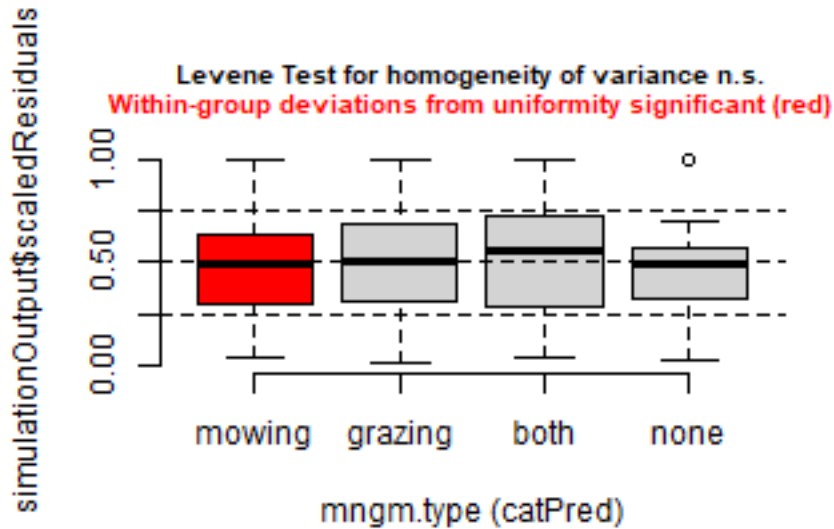
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
##   method           from
##   na.action.merMod lme4

##              GVIF Df GVIF^(1/(2*Df))
## esy4           3.025606 1      1.739427
## site.type      1.976839 2      1.185749
## eco.id         1.769052 2      1.153281
## obs.year       1.046940 1      1.023201
## esy4:site.type 3.250758 2      1.342753
## esy4:eco.id    2.248951 1      1.499650
```

```
car::vif(m_2)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## esy4           5.540917 1      2.353915
## site.type      59.153304 2      2.773287
## eco.id         21.268885 2      2.147515
## obs.year       6.206095 1      2.491204
## esy4:site.type  9.398477 2      1.750912
## esy4:eco.id     2.585528 1      1.607958
## esy4:obs.year   2.998932 1      1.731743
## site.type:eco.id 77.800058 4      1.723346
## site.type:obs.year 9.600051 2      1.760226
## eco.id:obs.year 11.673695 2      1.848426
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##           R2m           R2c
## [1,] 0.4169676 0.715171
MuMIn::r.squaredGLMM(m_2)
##           R2m           R2c
## [1,] 0.4546629 0.7235414
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnham & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_1 12 -580.7186
## m_2 21 -570.2147
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##              Chisq Df Pr(>Chisq)
## (Intercept)  692.5248  1 < 2.2e-16 ***
## esy4         68.3004  1 < 2.2e-16 ***
## site.type    2.6087  2  0.2713428
## eco.id       15.6626  2  0.0003971 ***
## obs.year     0.2986  1  0.5847781
## esy4:site.type 1.5111  2  0.4697597
## esy4:eco.id   13.2930  1  0.0002664 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_1)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: y ~ esy4 * (site.type + eco.id) + obs.year + (1 | id.site)
## Data: sites
##
##           AIC          BIC      logLik -2*log(L)  df.resid
##      -581.8      -537.9       302.9     -605.8       277
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.08092 -0.48018 -0.01625  0.41035  3.04610
##
## Random effects:
## Groups Name Variance Std.Dev.
## id.site (Intercept) 0.004589 0.06774
## Residual 0.004383 0.06620
## Number of obs: 289, groups: id.site, 118
```



```
##
## Fixed effects:
##           Estimate Std. Error t value
## (Intercept)      0.565453   0.021487  26.316
## esy4R1A          -0.230425   0.027882  -8.264
## site.type.L       0.038627   0.025364   1.523
## site.type.Q       0.012921   0.016537   0.781
## eco.id664        -0.087607   0.022400  -3.911
## eco.id686        -0.062701   0.022091  -2.838
## obs.year2023     -0.008427   0.015422  -0.546
## esy4R1A:site.type.L 0.055598   0.045928   1.211
## esy4R1A:site.type.Q 0.008598   0.030999   0.277
## esy4R1A:eco.id686  0.130702   0.035848   3.646
##
## Correlation of Fixed Effects:
##           (Intr) es4R1A st.t.L st.t.Q ec.664 ec.686 o.2023 e4R1A:...L
## esy4R1A      -0.641
## site.type.L -0.010 -0.042
## site.type.Q  0.447 -0.324  0.165
## eco.id664    -0.671  0.469  0.131 -0.063
## eco.id686    -0.632  0.458  0.056 -0.025  0.602
## obs.yer2023 -0.413  0.093  0.114 -0.018  0.073  0.003
## esy4R1A:...L -0.063  0.253 -0.511 -0.122 -0.041 -0.010  0.019
## esy4R1A:...Q -0.227  0.480 -0.119 -0.493  0.036  0.022  0.002  0.391
## es4R1A:.686  0.347 -0.506 -0.043  0.014 -0.341 -0.537  0.026  0.170
##           e4R1A:...Q
## esy4R1A
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## esy4R1A:...L
## esy4R1A:...Q
## es4R1A:.686  0.136
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

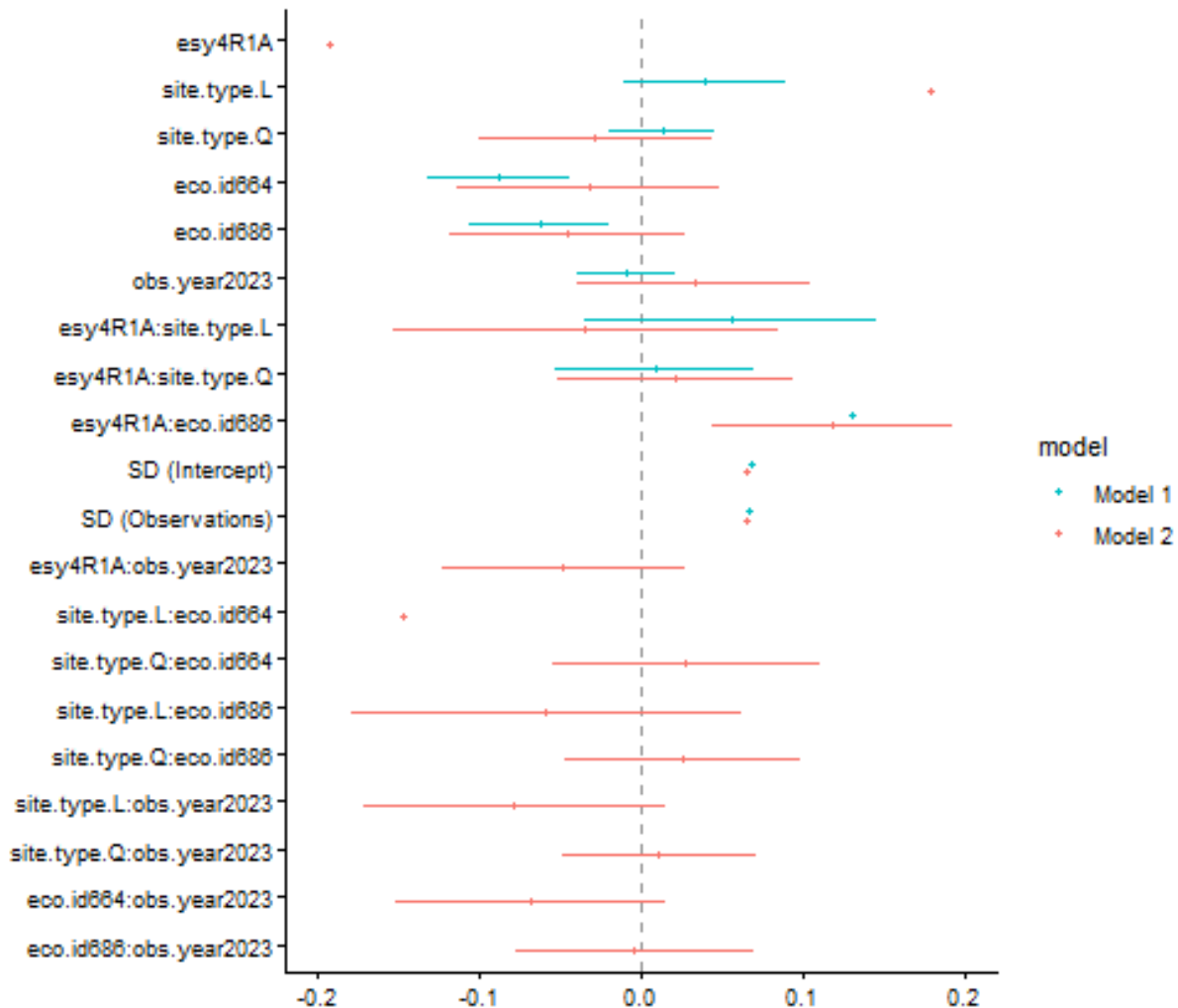
Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-0.2, 0.2) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```

```
## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
```

```
## i The deprecated feature was likely used in the dotwhisker package.
## Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_1,
  revpairwise ~ esy4,
  type = "response"
))
```

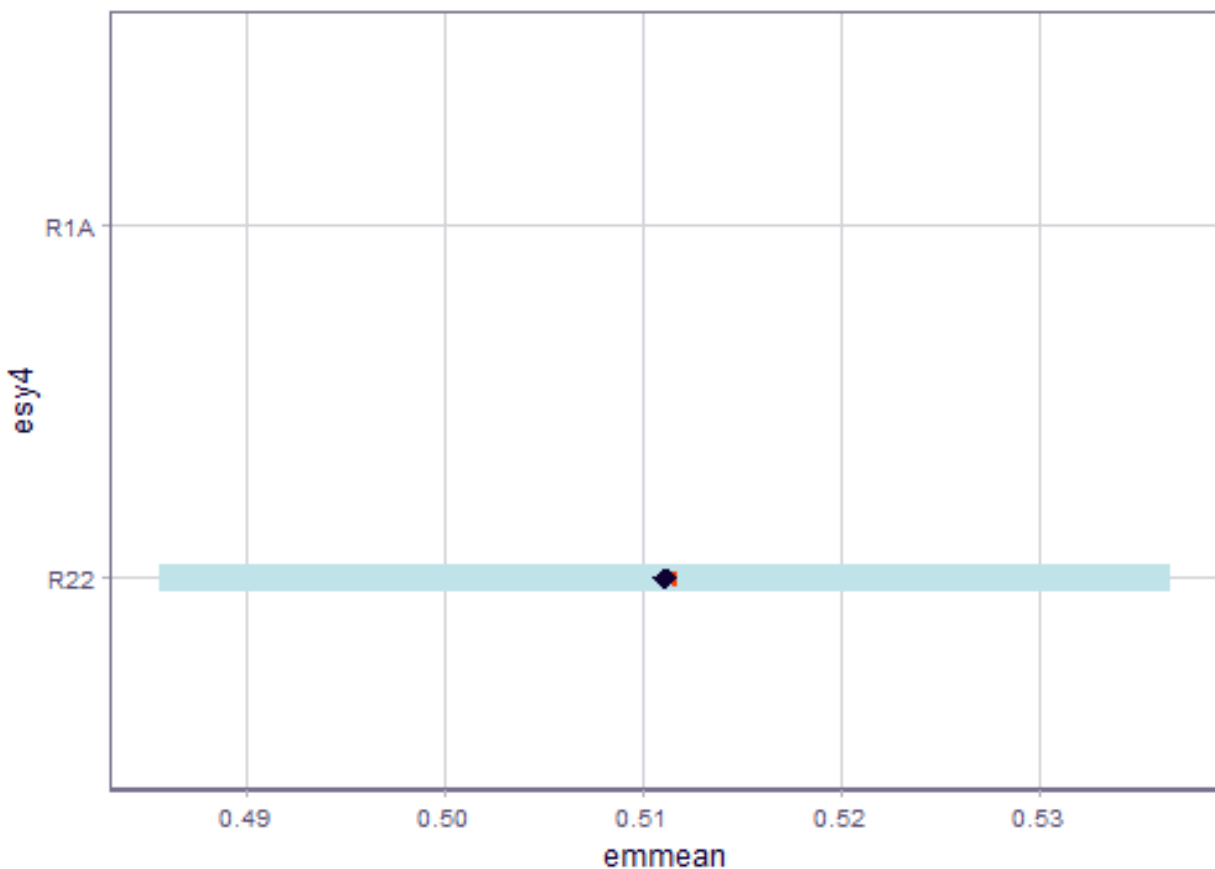
ESY EUNIS Habitat type

```
## $emmeans
## esy4 emmean SE df lower.CL upper.CL
```

```
## R22 0.511 0.0129 159 0.486 0.537
## R1A nonEst NA NA NA NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df z.ratio p.value
## R1A - R22 nonEst NA NA NA NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
```

```
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



```
(emm <- emmeans(
```

```
m_1,
  revpairwise ~ eco.id | esy4,
  type = "response"
))
```

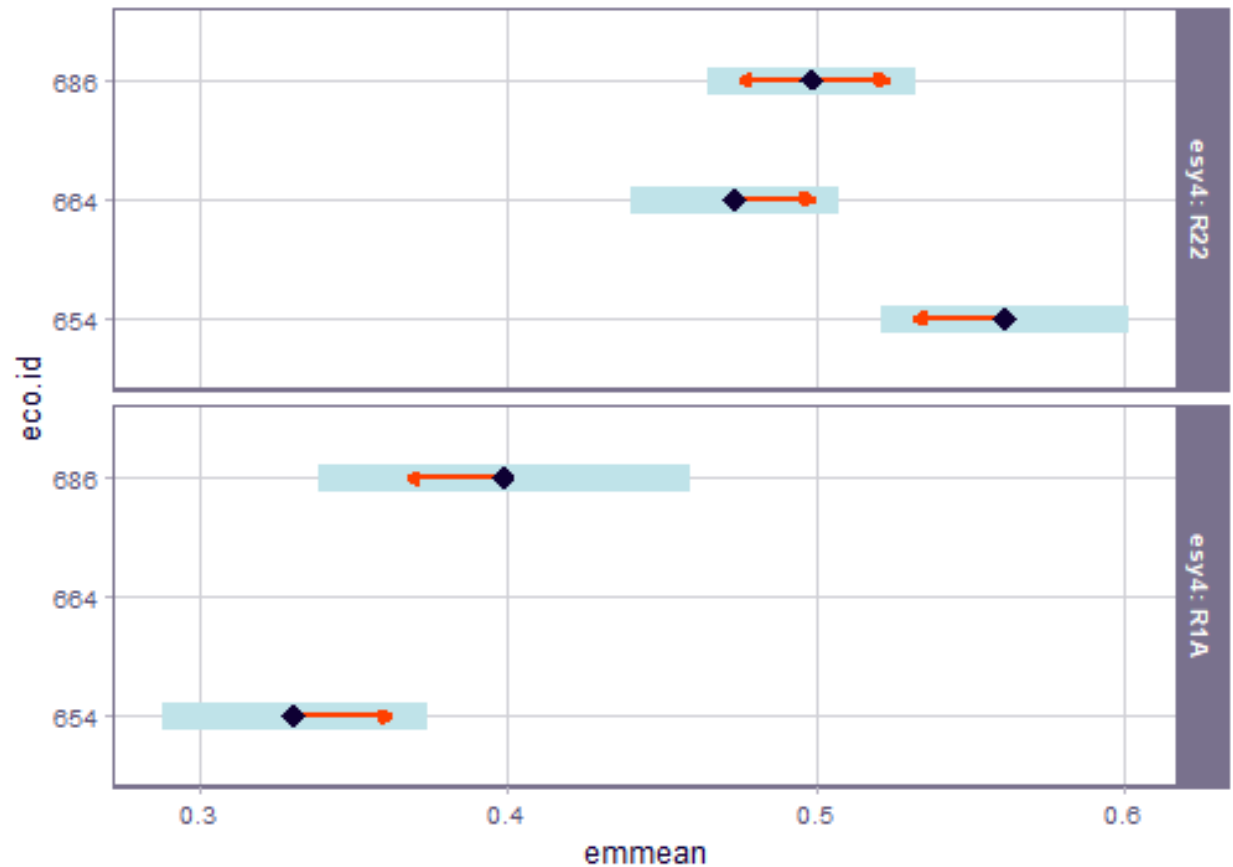
Habitat type x Region

```
## $emmeans
## esy4 = R22:
##   eco.id emmean      SE  df lower.CL upper.CL
##   654      0.561 0.0204 160    0.521    0.601
##   664      0.474 0.0170 130    0.440    0.507
##   686      0.499 0.0172 162    0.465    0.532
##
## esy4 = R1A:
##   eco.id emmean      SE  df lower.CL upper.CL
##   654      0.331 0.0217 129    0.288    0.374
##   664    nonEst      NA  NA      NA      NA
##   686      0.399 0.0305 158    0.339    0.459
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## esy4 = R22:
##   contrast                estimate      SE  df t.ratio p.value
##   eco.id664 - eco.id654   -0.0876 0.0233 140   -3.756 0.0007
##   eco.id686 - eco.id654   -0.0627 0.0230 154   -2.730 0.0193
##   eco.id686 - eco.id664    0.0249 0.0207 137    1.205 0.4520
##
## esy4 = R1A:
##   contrast                estimate      SE  df t.ratio p.value
##   eco.id664 - eco.id654    nonEst      NA  NA      NA      NA
##   eco.id686 - eco.id654    0.0680 0.0316 165    2.149 0.0331
##   eco.id686 - eco.id664    nonEst      NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for varying family sizes
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_2.0.3    DHARMa_0.5.0    patchwork_1.3.2 ggbeeswarm_0.7.3
## [5] lubridate_1.9.5  forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
## [9] purrr_1.2.2      readr_2.2.0     tidyr_1.3.2     tibble_3.3.1
```

```

## [13] ggplot2_4.0.3    tidyverse_2.0.0  here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.6      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.2.0       magrittr_2.0.5     multcomp_1.4-30
## [7] otel_0.2.0        compiler_4.5.2     mgcv_1.9-3
## [10] vctr_0.7.3        pkgconfig_2.0.3    crayon_1.5.3
## [13] fastmap_1.2.0     backports_1.5.1    labeling_0.4.3
## [16] utf8_1.2.6        ggstance_0.3.7     promises_1.5.0
## [19] rmarkdown_2.31    tzdb_0.5.0         nloptr_2.2.1
## [22] bit_4.6.0         xfun_0.58          later_1.4.8
## [25] broom_1.0.13      parallel_4.5.2     R6_2.6.1
## [28] gap.datasets_0.0.6 stringi_1.8.7       qgam_2.0.0
## [31] RColorBrewer_1.1-3 car_3.1-5           boot_1.3-32
## [34] estimability_1.5.1 Rcpp_1.1.1-1.1     iterators_1.0.14
## [37] knitr_1.51        zoo_1.8-15         parameters_0.29.1
## [40] httpuv_1.6.17     Matrix_1.7-4       splines_4.5.2
## [43] timechange_0.4.0  tidysselect_1.2.1  rstudioapi_0.18.0
## [46] abind_1.4-8       yaml_2.3.12        MuMIn_1.48.19
## [49] doParallel_1.0.17 codetools_0.2-20   lattice_0.22-7
## [52] plyr_1.8.9        bayestestR_0.18.1  shiny_1.13.0
## [55] withr_3.0.2       S7_0.2.2           evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3     pillar_1.11.1
## [61] gap_1.15.2        carData_3.0-6      foreach_1.5.2
## [64] stats4_4.5.2      insight_1.5.1      reformulas_0.4.4
## [67] generics_0.1.4    vroom_1.7.1        rprojroot_2.1.1
## [70] hms_1.1.4         scales_1.4.0       minqa_1.2.8
## [73] xtable_1.8-8      glue_1.8.1         tools_4.5.2
## [76] data.table_1.18.4 lme4_2.0-1         mvtnorm_1.4-0
## [79] grid_4.5.2        rbibutils_2.4.1    datawizard_1.3.1
## [82] nlme_3.1-168      Rmisc_1.5.1        performance_0.17.0
## [85] beeswarm_0.4.0    vipor_0.4.7        Formula_1.2-5
## [88] cli_3.6.6         gtable_0.3.6       digest_0.6.39
## [91] pbkrtest_0.5.5    TH.data_1.1-5      farver_2.1.2
## [94] htmltools_0.5.9   lifecycle_1.0.5    mime_0.13
## [97] bit64_4.8.2       dotwhisker_0.8.4   MASS_7.3-65

```